SOLVED PRACTICE WORKBOOK

Nuclear Fusion and ITER - Solved Practice Workbook

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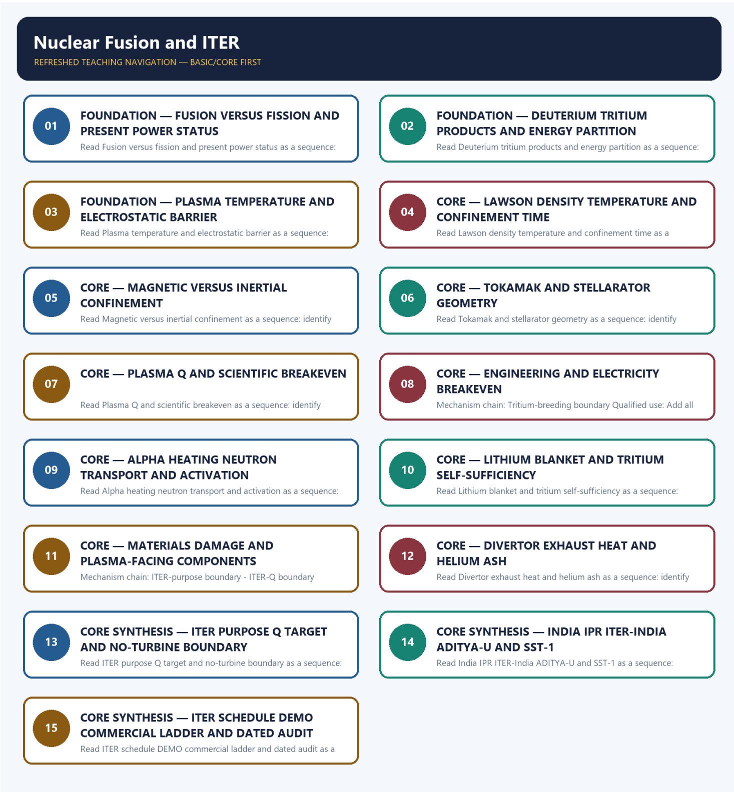

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Nuclear Fusion and ITER - Solved Practice Workbook

BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Fusion-fission boundary?

A. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. B. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. C. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. D. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating.

Answer: A. Explanation: Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q2. Which option preserves the technical boundary of Fusion-fission boundary?

A. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. B. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. C. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. D. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes.

Answer: B. Explanation: Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q3. Which statement uses Fusion-fission boundary without changing its institution, unit or status?

A. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. B. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. C. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. D. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant.

Answer: C. Explanation: Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q4. Which option avoids the standard UPSC close-option trap about Fusion-fission boundary?

A. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. B. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. C. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. D. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental.

Answer: D. Explanation: Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q5. Which statement correctly identifies DT-reaction boundary?

A. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. B. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. C. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. D. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition.

Answer: A. Explanation: The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q6. Which option preserves the technical boundary of DT-reaction boundary?

A. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. B. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. C. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. D. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes.

Answer: B. Explanation: The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q7. Which statement uses DT-reaction boundary without changing its institution, unit or status?

A. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. B. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. C. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. D. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes.

Answer: C. Explanation: The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q8. Which option avoids the standard UPSC close-option trap about DT-reaction boundary?

A. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. B. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. C. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. D. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating.

Answer: D. Explanation: The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q9. Which statement correctly identifies Temperature-plasma boundary?

A. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. B. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. C. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. D. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition.

Answer: A. Explanation: Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q10. Which option preserves the technical boundary of Temperature-plasma boundary?

A. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. B. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. C. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. D. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity.

Answer: B. Explanation: Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q11. Which statement uses Temperature-plasma boundary without changing its institution, unit or status?

A. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. B. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. C. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. D. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity.

Answer: C. Explanation: Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q12. Which option avoids the standard UPSC close-option trap about Temperature-plasma boundary?

A. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. B. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. C. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. D. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness.

Answer: D. Explanation: Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. The

other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q13. Which statement correctly identifies Lawson boundary?

A. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. B. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. C. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. D. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes.

Answer: A. Explanation: The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q14. Which option preserves the technical boundary of Lawson boundary?

A. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. B. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. C. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. D. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant.

Answer: B. Explanation: The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q15. Which statement uses Lawson boundary without changing its institution, unit or status?

A. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. B. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. C. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. D. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity.

Answer: C. Explanation: The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q16. Which option avoids the standard UPSC close-option trap about Lawson boundary?

A. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. B. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. C. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. D. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition.

Answer: D. Explanation: The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q17. Which statement correctly identifies Magnetic-inertial boundary?

A. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. B. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. C. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. D. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity.

Answer: A. Explanation: Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q18. Which option preserves the technical boundary of Magnetic-inertial boundary?

A. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. B. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. C. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. D. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity.

Answer: B. Explanation: Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q19. Which statement uses Magnetic-inertial boundary without changing its institution, unit or status?

A. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. B. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. C. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. D. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer.

Answer: C. Explanation: Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q20. Which option avoids the standard UPSC close-option trap about Magnetic-inertial boundary?

A. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. B. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. C. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. D. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes.

Answer: D. Explanation: Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q21. Which statement correctly identifies Tokamak-stellarator boundary?

A. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. B. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. C. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. D. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures.

Answer: A. Explanation: A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q22. Which option preserves the technical boundary of Tokamak-stellarator boundary?

A. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. B. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. C. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. D. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle.

Answer: B. Explanation: A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q23. Which statement uses Tokamak-stellarator boundary without changing its institution, unit or status?

A. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. B. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. C. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. D. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures.

Answer: C. Explanation: A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q24. Which option avoids the standard UPSC close-option trap about Tokamak-stellarator boundary?

A. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. B. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. C. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. D. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant.

Answer: D. Explanation: A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q25. Which statement correctly identifies Q-plasma boundary?

A. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. B. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. C. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. D. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer.

Answer: A. Explanation: Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q26. Which option preserves the technical boundary of Q-plasma boundary?

A. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. B. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. C. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. D. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement.

Answer: B. Explanation: Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q27. Which statement uses Q-plasma boundary without changing its institution, unit or status?

A. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. B. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. C. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. D. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle.

Answer: C. Explanation: Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q28. Which option avoids the standard UPSC close-option trap about Q-plasma boundary?

A. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. B. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. C. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. D. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity.

Answer: D. Explanation: Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q29. Which statement correctly identifies Engineering-breakeven boundary?

A. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. B. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. C. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. D. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement.

Answer: A. Explanation: Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q30. Which option preserves the technical boundary of Engineering-breakeven boundary?

A. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. B. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. C. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. D. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle.

Answer: B. Explanation: Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q31. Which statement uses Engineering-breakeven boundary without changing its institution, unit or status?

A. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. B. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. C. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. D. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant.

Answer: C. Explanation: Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. The other options change the scientific category, institutional owner, measurement,

mission rung or operating status.

Q32. Which option avoids the standard UPSC close-option trap about Engineering-breakeven boundary?

A. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. B. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. C. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. D. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer.

Answer: D. Explanation: Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q33. Which statement correctly identifies Alpha-neutron boundary?

A. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. B. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. C. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. D. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle.

Answer: A. Explanation: Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q34. Which option preserves the technical boundary of Alpha-neutron boundary?

A. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. B. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. C. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. D. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints.

Answer: B. Explanation: Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q35. Which statement uses Alpha-neutron boundary without changing its institution, unit or status?

A. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. B. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. C. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. D. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant.

Answer: C. Explanation: Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q36. Which option avoids the standard UPSC close-option trap about Alpha-neutron boundary?

A. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. B. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. C. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. D. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures.

Answer: D. Explanation: Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q37. Which statement correctly identifies Tritium-breeding boundary?

A. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. B. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. C. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. D. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant.

Answer: A. Explanation: Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q38. Which option preserves the technical boundary of Tritium-breeding boundary?

A. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. B. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. C. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. D. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints.

Answer: B. Explanation: Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q39. Which statement uses Tritium-breeding boundary without changing its institution, unit or status?

A. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. B. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. C. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. D. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims.

Answer: C. Explanation: Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q40. Which option avoids the standard UPSC close-option trap about Tritium-breeding boundary?

A. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. B. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. C. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. D. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle.

Answer: D. Explanation: Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q41. Which statement correctly identifies Materials boundary?

A. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. B. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. C. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. D. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints.

Answer: A. Explanation: High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q42. Which option preserves the technical boundary of Materials boundary?

A. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. B. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. C. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. D. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity.

Answer: B. Explanation: High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q43. Which statement uses Materials boundary without changing its institution, unit or status?

A. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. B. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. C. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. D. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms.

Answer: C. Explanation: High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q44. Which option avoids the standard UPSC close-option trap about Materials boundary?

A. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. B. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. C. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. D. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement.

Answer: D. Explanation: High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q45. Which statement correctly identifies Divertor boundary?

A. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. B. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. C. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. D. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims.

Answer: A. Explanation: A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q46. Which option preserves the technical boundary of Divertor boundary?

A. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. B. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. C. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. D. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity.

Answer: B. Explanation: A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q47. Which statement uses Divertor boundary without changing its institution, unit or status?

A. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. B. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. C. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. D. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims.

Answer: C. Explanation: A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q48. Which option avoids the standard UPSC close-option trap about Divertor boundary?

A. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. B. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. C. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. D. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints.

Answer: D. Explanation: A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q49. Which statement correctly identifies ITER-purpose boundary?

A. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. B. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. C. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. D. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms.

Answer: A. Explanation: ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q50. Which option preserves the technical boundary of ITER-purpose boundary?

A. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. B. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. C. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. D. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms.

Answer: B. Explanation: ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. The other options change the scientific category, institutional owner, measurement, mission rung or operating

status.

Q51. Which statement uses ITER-purpose boundary without changing its institution, unit or status?

A. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. B. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. C. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. D. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms.

Answer: C. Explanation: ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q52. Which option avoids the standard UPSC close-option trap about ITER-purpose boundary?

A. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. B. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. C. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. D. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant.

Answer: D. Explanation: ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q53. Which statement correctly identifies ITER-Q boundary?

A. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. B. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. C. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. D. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation.

Answer: A. Explanation: ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q54. Which option preserves the technical boundary of ITER-Q boundary?

A. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. B. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. C. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. D. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation.

Answer: B. Explanation: ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q55. Which statement uses ITER-Q boundary without changing its institution, unit or status?

A. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. B. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. C. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. D. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations.

Answer: C. Explanation: ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q56. Which option avoids the standard UPSC close-option trap about ITER-Q boundary?

A. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. B. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. C. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. D. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity.

Answer: D. Explanation: ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q57. Which statement correctly identifies Membership-contribution boundary?

A. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. B. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. C. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. D. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations.

Answer: A. Explanation: ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q58. Which option preserves the technical boundary of Membership-contribution boundary?

A. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. B. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. C. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. D. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations.

Answer: B. Explanation: ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q59. Which statement uses Membership-contribution boundary without changing its institution, unit or status?

A. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. B. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. C. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. D. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries.

Answer: C. Explanation: ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q60. Which option avoids the standard UPSC close-option trap about Membership-contribution boundary?

A. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. B. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. C. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. D. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims.

Answer: D. Explanation: ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q61. Which statement correctly identifies IPR-ITERIndia boundary?

A. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. B. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. C. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. D. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation.

Answer: A. Explanation: IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q62. Which option preserves the technical boundary of IPR-ITERIndia boundary?

A. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. B. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. C. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. D. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost.

Answer: B. Explanation: IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q63. Which statement uses IPR-ITERIndia boundary without changing its institution, unit or status?

A. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. B. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. C. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. D. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost.

Answer: C. Explanation: IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q64. Which option avoids the standard UPSC close-option trap about IPR-ITERIndia boundary?

A. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. B. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. C. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. D. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms.

Answer: D. Explanation: IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q65. Which statement correctly identifies Domestic-device boundary?

A. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. B. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. C. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. D. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries.

Answer: A. Explanation: ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q66. Which option preserves the technical boundary of Domestic-device boundary?

A. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. B. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. C. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. D. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost.

Answer: B. Explanation: ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q67. Which statement uses Domestic-device boundary without changing its institution, unit or status?

A. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. B. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. C. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. D. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating.

Answer: C. Explanation: ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q68. Which option avoids the standard UPSC close-option trap about Domestic-device boundary?

A. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. B. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. C. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. D. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation.

Answer: D. Explanation: ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q69. Which statement correctly identifies Schedule-status boundary?

A. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. B. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. C. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. D. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries.

Answer: A. Explanation: ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q70. Which option preserves the technical boundary of Schedule-status boundary?

A. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. B. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. C. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. D. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries.

Answer: B. Explanation: ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q71. Which statement uses Schedule-status boundary without changing its institution, unit or status?

A. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. B. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. C. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. D. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness.

Answer: C. Explanation: ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q72. Which option avoids the standard UPSC close-option trap about Schedule-status boundary?

A. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. B. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. C. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. D. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations.

Answer: D. Explanation: ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q73. Which statement correctly identifies DEMO-commercial boundary?

A. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. B. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. C. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. D. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental.

Answer: A. Explanation: A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q74. Which option preserves the technical boundary of DEMO-commercial boundary?

A. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. B. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. C. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. D. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating.

Answer: B. Explanation: A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q75. Which statement uses DEMO-commercial boundary without changing its institution, unit or status?

A. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. B. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. C. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. D. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness.

Answer: C. Explanation: A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q76. Which option avoids the standard UPSC close-option trap about DEMO-commercial boundary?

A. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. B. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. C. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. D. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost.

Answer: D. Explanation: A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q77. Which statement correctly identifies Volatile fusion boundary?

A. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. B. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. C. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. D. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental.

Answer: A. Explanation: Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain

plasma-versus-plant boundaries. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q78. Which option preserves the technical boundary of Volatile fusion boundary?

A. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. B. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. C. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. D. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating.

Answer: B. Explanation: Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q79. Which statement uses Volatile fusion boundary without changing its institution, unit or status?

A. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. B. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. C. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. D. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes.

Answer: C. Explanation: Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q80. Which option avoids the standard UPSC close-option trap about Volatile fusion boundary?

A. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. B. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. C. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. D. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries.

Answer: D. Explanation: Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

Audited ledgers route the direct 2025 GS-III ITER contribution and future-energy demand here. No other direct PYQ or answer key is manufactured.

PYQ DEMAND CARD 1 - 2025 GS-III

Demand: Mention India's contributions to ITER and explain the implications of project success for global energy.

Status: Verified direct routed Mains demand.

Model solution: Fusion-fission boundary: Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. **Q-plasma boundary:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Engineering-breakeven boundary:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Tritium-breeding boundary:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Materials boundary:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **ITER-purpose boundary:** ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. **ITER-Q boundary:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Membership-contribution boundary:** ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. **IPR-ITERIndia boundary:** IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. **Domestic-device boundary:** ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. **DEMO-commercial boundary:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Volatile fusion boundary:** Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: The directive **answer** requires a direct position on “PYQ DEMAND CARD 1 - 2025 GS-III”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Fusion-fission boundary: Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. **Q-plasma boundary:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Engineering-breakeven boundary:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Tritium-breeding boundary:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Materials boundary:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **ITER-purpose boundary:** ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. **ITER-Q boundary:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Membership-contribution boundary:** ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. **IPR-ITERIndia boundary:** IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. **Domestic-device boundary:** ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. **DEMO-commercial boundary:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Volatile fusion boundary:** Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Mention India's contributions to ITER and explain the implications of project success for global energy. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Fusion-fission boundary: Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. **Q-plasma boundary:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Engineering-breakeven boundary:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Tritium-breeding boundary:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Materials boundary:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **ITER-purpose boundary:** ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. **ITER-Q boundary:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Membership-contribution boundary:** ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. **IPR-ITERIndia boundary:** IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. **Domestic-device boundary:** ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. **DEMO-commercial boundary:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Volatile fusion boundary:** Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “PYQ DEMAND CARD 1 - 2025 GS-III”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 1 - 10 MARKS

Question: Explain why ITER must not be described as a commercial power plant. Answer in about 150 words.

Model thesis: Claim: Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-purpose boundary. **Named evidence/example:** ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-Q boundary. **Named evidence/example:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** DEMO-commercial boundary. **Named evidence/example:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity.
- Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer.
- ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant.
- ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity.
- A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost.

Qualified conclusion: Claim: Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-purpose boundary. **Named evidence/example:** ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-Q boundary. **Named evidence/example:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** DEMO-commercial boundary. **Named evidence/example:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **explain** requires a direct position on “Explain why ITER must not be described as a commercial power plant. Answer in about 150 words.”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-purpose boundary. **Named evidence/example:** ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-Q boundary. **Named evidence/example:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** DEMO-commercial boundary. **Named evidence/example:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. **Analysis:**

Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

4. **Claim and named evidence:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-purpose boundary. **Named evidence/example:** ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-Q boundary. **Named evidence/example:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** DEMO-commercial boundary. **Named evidence/example:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Explain why ITER must not be described as a commercial power plant. Answer in about 150 words.”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 2 - 10 MARKS

Question: Distinguish plasma Q from engineering and electricity breakeven. Answer in about 150 words.

Model thesis: Claim: Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-Q boundary. **Named evidence/example:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity.
- Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer.
- ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity.

Qualified conclusion: Claim: Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-Q boundary. **Named evidence/example:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **answer** requires a direct position on “Distinguish plasma Q from engineering and electricity breakeven. Answer in about 150 words.”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-Q boundary. **Named evidence/example:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-Q boundary. **Named evidence/example:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Distinguish plasma Q from engineering and electricity breakeven. Answer in about 150 words.”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 3 - 15 MARKS

Question: Explain the D-T fusion chain and its tritium and materials constraints. Answer in about 250 words.

Model thesis: Claim: DT-reaction boundary. **Named evidence/example:** The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Temperature-plasma boundary. **Named evidence/example:** Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Lawson boundary. **Named evidence/example:** The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Alpha-neutron boundary. **Named evidence/example:** Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tritium-breeding boundary. **Named evidence/example:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Materials boundary. **Named evidence/example:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Divertor boundary. **Named evidence/example:** A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating.
- Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness.
- The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition.
- Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures.
- Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle.
- High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement.
- A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints.

Qualified conclusion: Claim: DT-reaction boundary. **Named evidence/example:** The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Temperature-plasma boundary. **Named evidence/example:** Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Lawson boundary. **Named evidence/example:** The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Alpha-neutron boundary. **Named evidence/example:** Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tritium-breeding boundary. **Named evidence/example:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Materials boundary. **Named evidence/example:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Divertor boundary. **Named evidence/example:** A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **explain** requires a direct position on “Explain the D-T fusion chain and its tritium and materials constraints. Answer in about 250…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: DT-reaction boundary. **Named evidence/example:** The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Temperature-plasma boundary. **Named evidence/example:** Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Lawson boundary. **Named evidence/example:** The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Alpha-neutron boundary. **Named evidence/example:** Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1

MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tritium-breeding boundary. **Named evidence/example:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Materials boundary. **Named evidence/example:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Divertor boundary. **Named evidence/example:** A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
7. **Claim and named evidence:** A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. **Analysis:** Connect

mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: DT-reaction boundary. **Named evidence/example:** The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Temperature-plasma boundary. **Named evidence/example:** Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Lawson boundary. **Named evidence/example:** The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Alpha-neutron boundary. **Named evidence/example:** Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tritium-breeding boundary. **Named evidence/example:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Materials boundary. **Named evidence/example:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Divertor boundary. **Named evidence/example:** A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Explain the D-T fusion chain and its tritium and materials constraints. Answer in about 250…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 4 - 15 MARKS

Question: Assess India's domestic fusion ecosystem and ITER participation. Answer in about 250 words.

Model thesis: Claim: Membership-contribution boundary. **Named evidence/example:** ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IPR-ITERIndia boundary. **Named evidence/example:** IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Domestic-device boundary. **Named evidence/example:** ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile fusion boundary. **Named evidence/example:** Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims.
- IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms.
- ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation.
- Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries.

Qualified conclusion: Claim: Membership-contribution boundary. **Named evidence/example:** ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IPR-ITERIndia boundary. **Named evidence/example:** IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Domestic-device boundary. **Named evidence/example:** ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile fusion boundary. **Named evidence/example:** Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **assess** requires a direct position on "Assess India's domestic fusion ecosystem and ITER participation. Answer in about 250 words.", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Membership-contribution boundary. **Named evidence/example:** ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IPR-ITERIndia boundary. **Named evidence/example:** IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Domestic-device boundary. **Named evidence/example:** ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile fusion boundary. **Named evidence/example:** Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Membership-contribution boundary. **Named evidence/example:** ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IPR-ITERIndia boundary. **Named evidence/example:** IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Domestic-device boundary. **Named evidence/example:** ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile fusion boundary. **Named evidence/example:** Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Assess India's domestic fusion ecosystem and ITER participation. Answer in about 250 words.”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 5 - 20 MARKS

Question: Evaluate fusion through scientific, engineering and commercial feasibility thresholds. Answer in about 300 words.

Model thesis: Claim: Temperature-plasma boundary. **Named evidence/example:** Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Lawson boundary. **Named evidence/example:** The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Magnetic-inertial boundary. **Named evidence/example:** Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tokamak-stellarator boundary. **Named evidence/example:** A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tritium-breeding boundary. **Named evidence/example:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Materials boundary. **Named evidence/example:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Divertor boundary. **Named evidence/example:** A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** DEMO-commercial boundary. **Named evidence/example:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness.
- The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition.
- Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes.
- A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant.
- Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity.
- Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer.
- Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle.
- High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement.
- A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints.
- A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost.

Qualified conclusion: Claim: Temperature-plasma boundary. **Named evidence/example:** Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Lawson boundary. **Named evidence/example:** The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Magnetic-inertial boundary. **Named evidence/example:** Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tokamak-stellarator boundary. **Named evidence/example:** A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tritium-breeding boundary. **Named evidence/example:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Materials boundary. **Named evidence/example:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Divertor boundary. **Named evidence/example:** A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** DEMO-commercial boundary. **Named evidence/example:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **evaluate** requires a direct position on “Evaluate fusion through scientific, engineering and commercial feasibility thresholds. Answer…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Temperature-plasma boundary. **Named evidence/example:** Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Lawson boundary. **Named evidence/example:** The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Magnetic-inertial boundary. **Named evidence/example:** Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tokamak-stellarator boundary. **Named evidence/example:** A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tritium-breeding boundary. **Named evidence/example:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Materials boundary. **Named evidence/example:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Divertor boundary. **Named evidence/example:** A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** DEMO-commercial boundary. **Named evidence/example:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. **Analysis:** Connect

mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

3. **Claim and named evidence:** Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
7. **Claim and named evidence:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
8. **Claim and named evidence:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
9. **Claim and named evidence:** A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
10. **Claim and named evidence:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Temperature-plasma boundary. **Named evidence/example:** Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Lawson boundary. **Named evidence/example:** The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Magnetic-inertial boundary. **Named evidence/example:** Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tokamak-stellarator boundary. **Named evidence/example:** A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tritium-breeding boundary. **Named evidence/example:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Materials boundary. **Named evidence/example:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Divertor boundary. **Named evidence/example:** A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** DEMO-commercial boundary. **Named evidence/example:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Evaluate fusion through scientific, engineering and commercial feasibility thresholds. Answer…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 6 - 20 MARKS

Question: Analyse the implications of ITER success without overstating timelines or electricity output. Answer in about 300 words.

Model thesis: Claim: Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tritium-breeding boundary. **Named evidence/example:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Materials boundary. **Named evidence/example:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-purpose boundary. **Named evidence/example:** ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; 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Claim -> named evidence -> analysis -> qualification:

- Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity.
- Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer.
- Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle.
- High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement.
- ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant.
- ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity.
- ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims.
- IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms.
- ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations.
- A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost.
- Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries.

Qualified conclusion: Claim: Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tritium-breeding boundary. **Named evidence/example:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; 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Demand decoding: The directive **analyse** requires a direct position on “Analyse the implications of ITER success without overstating timelines or electricity output.…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tritium-breeding boundary. **Named evidence/example:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; 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Analytical body:

1. **Claim and named evidence:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
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4. **Claim and named evidence:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
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9. **Claim and named evidence:** ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
10. **Claim and named evidence:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
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Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tritium-breeding boundary. **Named evidence/example:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; 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Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Analyse the implications of ITER success without overstating timelines or electricity output.…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.